MIL-P-22299B(SH)
15 June 1985
SUPERSEDING
MIL-P-22299A(SHIPS)
27 November 1963
(See 6.9)

MILITARY SPECIFICATION

PAINT, ANTIFOULING, POLYISOBUTYLENE (FORMULA NO. 134)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polyisobutylene antifouling paint for use over underwater rubber surfaces painted with black polyisobutylene paint (formula no. 133).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-N-95	- Naphtha, Aliphatic.
TT-T-291	- Thinner, Paint, Mineral Spirits, Regular and Odorless.
TT-T-548	- Toluene, Technical.
LLL-R-626	- Rosins: Gum, Wood, and Tall Oil.
PPP-P-1892	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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MILITARY

MIL-P-15169 - Pigment, Cuprous Oxide.
MIL-P-22298 - Paint, Black, Polyisobutylene (Formula No. 133).

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods for Sampling and Testing.
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 562 - Standard Test Methods for Consistency of Paints Using the Stormer Viscometer. (DoD adopted)
D 846 - Standard Specification for Ten-Degree Xylene. (DoD adopted)
D 1153 - Standard Specification for Methyl Isobutyl Ketone.
D 1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems. (DoD adopted)
D 1296 - Standard Test Method for Odor of Volatile Solvents and Diluents. (DoD adopted)
D 1475 - Standard Test Method for Density of Paint, Varnish, Lacquer, and Related Products. (DoD adopted)
D 1542 - Standard Test Method for Qualitative Tests for Rosin in Varnishes. (DoD adopted)
D 1545 - Standard Test Method for Viscosity of Transparent Liquids by Bubble Time Method. (DoD adopted)
D 1729 - Standard Practice for Visual Evaluation of Color Differences of Opaque Materials. (DoD adopted)
D 2369 - Standard Test Method for Volatile Content of Coatings. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rules and Regulations
Rule 102

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Formula.

3.1.1 The paint shall consist of ingredients, conforming to the applicable specifications, in the proportions specified in table I.

TABLE I. Formula no. 134.

Ingredients	Applicable specifications	Pounds per 100 gallons 1/
Cuprous oxide	MIL-P-15169	590.0
Rosin	LLL-R-626, class A, grade WW or WG, type I	27.0
Polyisobutylene polymer 2/	---	38.0
Mineral spirits	TT-T-291, type II	490.0
Toluene	TT-T-548	70.0
Inhibitor (di-tert-butyl-para-cresol) 3/	---	0.05

1/ The formula is slightly in excess of 100 gallons to allow for normal manufacturing loss.

2/ The polyisobutylene polymer shall be composed entirely of isobutylene polymers (a polymer having a Staudinger molecular weight of approximately 100,000 should be satisfactory). A 7.5 percent solution of this polymer in mineral spirits conforming to type II of TT-T-291 shall have a Gardner-Holdt viscosity between X and Z3 when tested in accordance with ASTM D 1545.

3/ Naugard BHT of Uniroyal, DBPC of Koppers Company, CAO-1 of Ashland Chemical Company, or equivalent. The formula specified in table I is designated as Navy standard formula no. 134. When formula no. 134 is specified, the paint shall conform to this specification.

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3.2 Manufacture. The component raw materials of the paints shall be mixed and ground as required to produce a product which is uniform, stable, free from grit, suitable for the purpose intended, and in full conformity to the requirements of this specification.

3.3 Volatile portion. The volatile portion of the enamel shall conform to the following requirements by volume (see 4.4):

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene, methyl benzoate and phenyl acetate: 8 percent maximum.
(b) Ethyl benzene, toluene and branched chain ketones: 20 percent maximum.
(c) Compounds with olefinic or cyclo-olefinic type of unsaturation: 6 percent maximum.
(d) Total (a), (b), and (c): 20 percent maximum.

3.4 Quantitative requirements. The paint shall be as specified in table II, when tested in accordance with table III.

TABLE II. Quantitative requirements.

Characteristics	Minimum	Maximum
Pigment, percent by weight	47.0	51.0
Volatiles, percent by weight	44.0	48.0
Nonvolatile vehicle, percent by weight (calculated by difference)	4.9	5.7
Water, percent by weight	----	0.5
Coarse particles and skins, as residue retained on no. 325 sieve, percent by weight	----	0.5
Consistency, Krebs-Stormer shearing rate 200 r/min		
Grams	175.0	335.0
Krebs unit (equivalent)	77.0	98.0
Fineness of grind	5.0	----
Weight per gallon, pounds	11.4	12.0
Dry hard time, minutes	----	90.0
Pigment composition: Total copper, percent by weight of pigment	86.0	----
Vehicle portion insoluble in methyl isobutyl ketone, percent by weight of nonvolatile vehicle	55.0	65.0

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3.5 Qualitative requirements. The paint shall conform to the qualitative requirements as specified in 3.5.1 through 3.5.7.

3.5.1 Odor. The odor shall be characteristic of the volatiles permitted.

3.5.2 Color. The color shall be characteristic of the pigment used.

3.5.3 Rosin. Rosin shall be present.

3.5.4 Surface appearance. The paint applied as specified in 4.3.7.2 shall be smooth and uniform, and shall show no evidence of cracking or other defects.

3.5.5 Adhesion. The paint shall adhere to black polyisobutylene paint in accordance with MIL-P-22298 when tested as specified in 4.3.7.2.

3.5.6 Elasticity. The paint shall show no cracking or loss of adhesion when stretched as specified in 4.3.7.3.

3.5.7 Storage stability. For at least 1 year after date of acceptance, the paint shall readily break up with a paddle to a smooth, uniform consistency; shall not exceed 110 Krebs units in viscosity; shall not exceed 2 hours dry hard time; shall not curdle, gel nor show any other objectionable properties such as pigment settling.

3.6 Hazardous materials. Lead, chromate or asbestos shall not be present.

3.7 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting activity.

3.8 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS is DD Form OSHA 20 and is found in FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance. Quality assurance inspection shall be as specified in table III.

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4.2.1 Inspection lot. A lot shall consist of paint of the same type, class and grade from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.2.2 Sampling plan. Two representative random samples from each lot shall be tested. The samples shall be packaged in separate containers. The minimum size for each sample shall be 1 quart.

4.3 Test procedures. The paint shall be tested in accordance with table III. Failure of any sample to pass any test and noncompliance with the requirements of this specification shall be cause for rejection of the lot represented by the sample.

TABLE III. Test procedures.

Tests	Applicable test method FED-STD-141	Applicable ASTM test method	Test paragraph	Requirement paragraph
Pigment content	1/ 4021	----	4.3.1	Table II
Volatiles	----	D 2369	----	Table II
Nonvolatile vehicle	4053	----	----	Table II
Water	4081	----	----	Table II
Coarse particles	1/ 4092	----	----	Table II
Consistency	----	D 562	----	Table II
Fineness of grind	----	D 1210	----	Table II
Weight per gallon	----	D 1475	----	Table II
Drying time	4061	----	4.3.2	Table II
Copper	----	----	4.3.3	Table II
Vehicle portion insoluble	----	----	4.3.4	Table II
Odor	----	D 1296	----	3.5.1
Color	----	D 1729	4.3.5	3.5.2
Rosin	----	D 1542	----	3.5.3
Surface appearance	----	----	4.3.6	3.5.4
Adhesion	----	----	4.3.7.2	3.5.5

See footnote at end of table.

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TABLE III. Test procedures. - Continued

Tests	Applicable test method FED-STD-141	Applicable ASTM test method	Test paragraph	Requirement paragraph
Elasticity	----	----	4.3.7.3	3.5.6
Condition in container	----	----	----	3.5.7

1/ Toluene shall be used as the extraction liquid.

4.3.1 Pigment content. The pigment content of the paint shall be determined in accordance with method 4021 of FED-STD-141 using toluene as the extraction liquid. The first three extractions shall be saved for the test specified in 4.3.4.

4.3.2 Drying time. The drying time shall be determined in accordance with method 4061 of FED-STD-141, except the wet film thickness of the paint drawn down shall be 3 mils.

4.3.3 Total copper. Total copper shall be determined as follows: Weigh accurately 1.0 gram of the extracted pigment and transfer to a 300-milliliter (mL) electrolytic beaker. Add 10 mL nitric acid (specific gravity 1.42) and boil for several minutes. (If solution is not clear, cool and dilute with 100 mL water, then filter and wash with hot water, hot 10 percent nitric acid, and finally with hot water.) Add 10 mL perchloric acid (70 percent) and fume for 5 minutes. Cool, dilute to a volume of approximately 150 mL, and add 1 mL of concentrated nitric acid and 1 gram of ammonium sulfate. Electroplate the copper on a rotating platinum electrode at 2 amperes current for 2 hours. Dilute with water and continue the electrolysis for 15 minutes. When the deposition is complete, remove electrode, wash with water and acetone, dry and weigh. Calculate the percentage of total copper.

4.3.4 Vehicle portion insoluble in methyl isobutyl ketone. Insoluble vehicle portion shall be determined as follows: Collect the decanted liquid from the first three extractions as specified in 4.3.1 and dilute with equal volume of methyl isobutyl ketone in accordance with ASTM D 1153. Filter the insoluble material through a weighted fritted glass filter. Wash with several 20 mL portions of a half and half mixture of methyl isobutyl ketone and xylene in accordance with ASTM D 846. Dry the fritted glass filter in 110-degrees Celsius (°C) oven for 3 hours and weigh. Calculate the insoluble vehicle portion as percent of the total nonvolatile vehicle.

4.3.5 Color. The color of the paint film prepared as specified in 4.3.7.2 shall be examined in accordance with 3.5.2 (see 6.6).

4.3.6 Surface appearance. The surface of the dry paint film prepared as specified in 4.3.7.2 shall be examined in accordance with 3.5.4.

4.3.7 Adhesion and elasticity. Adhesion and elasticity shall be determined as specified in 4.3.7.1 through 4.3.7.3 and with temperature and humidity in accordance with FED-STD-141.

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4.3.7.1 Composition of rubber on which the coating is applied shall be as specified in table IV.

TABLE IV. Rubber composition.

Materials	Parts by weight
Smoked sheet	100.0
Sulphur	3.0
Zinc oxide	5.0
Captax[1/] or equivalent (2-mercaptobenzothiazole)	0.5
Stearic acid	0.5
Helizone[2/] or equivalent (blend of petroleum waxes)	1.0
Neozone[1/] or equivalent (phenyl naphthylamine)	1.0
P33 black[2/] or equivalent	0.5

1/ R.T. Vanderbilt Co., Inc., 230 Park Avenue, New York, NY 10017.
2/ E.I. duPont de Nemours and Co., Wilmington, DE 19898.

The rubber shall be vulcanized in a 0.075 by 6 by 6 inch mold using a minimum pressure of 500 pounds per square inch (lb/in^2) at a temperature of 143°C for 15 minutes. Sample of rubber may be obtained from David Taylor Naval Ship Research and Development Center, Annapolis, MD 21402.

4.3.7.2 Adhesion. One surface of the test rubber shall be cleaned by wiping firmly and briskly with a small cotton wad moistened with aliphatic naphtha in accordance with TT-N-95. To minimize swelling of the rubber, the excess naphtha shall be pressed from the wad before wiping and no more than five strokes shall be taken over any single spot on the surface. The rubber surface shall not appear wet during wiping except behind the moving wad during each stroke. After cleaning the surface, 30 minutes shall be allowed for the naphtha to evaporate and the test rubber to resume a flat position. Three coats of the black polyisobutylene paint in accordance with MIL-P-22298 (1 to 2 mils wet thickness per coat) shall be applied to the test rubber. Dry time between coats shall be 3 hours minimum and 18 hours maximum. Two coats (each 2 to 3 mils wet thickness per coat) of the sample antifouling paint shall be applied under test, allowing 3 hours minimum and 18 hours maximum drying time between coats. Twenty hours after the application of the final coat and thereafter, the paint shall be as specified in 3.5.5. The paint shall show no signs of flaking or separation when pried with a knife or flat spatula.

4.3.7.3 Elasticity. A coated rubber panel in accordance with 4.3.7.1 and 4.3.7.2 shall be stretched uniformly to an additional 10 percent of its length, 20 hours after receiving the final coat. The paint shall adhere to the undercoat and show no sign of alligatoring or cracking when examined under five power magnification.

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4.4 Certification data/reports. The contractor shall certify that each lot of paint meets specification requirements in accordance with the data ordering document included in the contract (see 6.2.2). In addition to the requirements covered by the data ordering document, the certification data/report shall include the following information:

(a) Conformance of each lot of ingredient material to table I, stating test results and source, as applicable (see 4.2).
(b) Certification that volatility requirements are met (see 3.3).
(c) Conformance to specification requirements and test results as specified (see 4.3) for finished paint.

4.5 Toxicity. The contractor shall furnish to the contracting activity the toxicological data and formulations required to evaluate the safety of the material for the proposed use.

4.6 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing, and marking. The enamel shall be packaged, packed, and marked in accordance with PPP-P-1892. The level of packaging shall be level A, B or C, and the level of packing shall be level A, B or C as specified (see 6.2.1). The enamel shall be furnished in 1-gallon cans or 5-gallon pails as specified (see 6.2.1).

5.1.1 Special marking. In addition to other marking required on the containers there shall be the following statement: "The volatile content of this container is not photochemically reactive as defined by Rule 102 of the South Coast Air Quality Management District" (see 6.7).

6. NOTES

6.1 Intended use. This paint is intended for use as an antifouling paint for rubber surfaces exposed to seawater. It is applied over black polyisobutylene paint formula no. 133 in accordance with MIL-P-22298 when the rubber surface is of a type sensitive to copper ions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of container (see 5.1).
(c) Level of packaging and packing required (see 5.1).

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6.2.1.1 Level B packaging. Level B packaging is intended to provide economical but limited protection and should be specified only when it is determined the paint will be held in covered storage no more than 1 year from the date of initial packaging.

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52.227-7031 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4	Certification data/report	UDI-A-23264	10.2.1

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4 or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Paint should be purchased by volume, the unit being a U.S gallon at 60°F (15.5°C).

6.4 Patent notice. The Government has a royalty-free license under U.S patent number 3,033,809 for the benefit of manufacturers of the item either for the Government or for use to be delivered to the Government.

6.5 Composition by volume. For information only, and with the understanding that weight volume relationships of ingredients may vary slightly, the approximation of composition by volume is specified in table V.

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TABLE V. Composition by volume.

Ingredients	Gallons
Cuprous oxide	11.8
Rosin	3.0
Polyisobutylene polymer	4.8
Paint thinner	74.3
Toluene	9.7
Inhibitor	0.005
Total	103.605

6.6 The color of most formula no. 134 paint approximates color 30166 in accordance with FED-STD-595. However, the color of cuprous oxide pigment is not closely standardized. The color of a specific batch will depend upon the particular cuprous oxide used in manufacture.

6.7 Volatile content. Although the container marking specifically refers to the South Coast Air Quality Management District, the paints may be used anywhere else a product complying with 3.3 is allowed. This includes other air pollution control districts or similar areas controlling the emission of solvents into the atmosphere. Information regarding Rule 102 may be obtained from the South Coast Air Quality Management District.

6.8 Recommended manufacturing process. The polyisobutylene polymer (cut up into small pieces) and inhibitor should be added to nearly all the paint thinner to form a 7.5 percent solution. (Unlike rubber, polyisobutylene polymer should not be masticated before preparing the solution as this causes degradation with resulting deterioration of mechanical properties. If rolls are used to draw the polymer out in thin sheets before introduction into the paint thinner, mastication should be held to the least possible amount. Solution of the polymer is a slow swelling process, relatively independent of the rate of agitation. The minimum agitation which prevents settling and compaction of the polymer is recommended.) The cuprous oxide pigment should be mixed with a portion of the 7.5 percent polymer solution to make a mill paste, and the paste ground on a roller mill. The rosin should be dissolved in the rest of the paint thinner and the toluene. The mill paste, remaining polyisobutylene polymer solution, and rosin solution should be blended to form the finished paint.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-22299B(SH)	Paint, Antifouling, Polyisobutylene (Formula No. 134)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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